

```
1 C:\Users\13400f_RTX4060ti\AppData\Local\Programs\Python\
  Python313\python.exe D:\MoleColyte\train_Tox21.py
2 Splitting: Train: 4408 |Val: 551 | Test: 551
3 Kiln: cuda
4 Scanning training data to calculate exact imbalance penalties...
5 Calculated Pathway Penalties: [31.7 49.1 7.9 31.7 8.4 23.6 51.5 7.8 26.9
  21.7 7.9 20.1]
6
7 Starting Epoch 1
8   Epoch 1/20 | Batch 0 | Loss: 156.6682
9 ==> Epoch 1 Train Complete | Average BCE Loss: 10.9551
10  Validation Phase | Avg Val Loss: 1.6482
11  □ New best Tox21 model saved! (Lowest Val Loss: 1.6482)
12
13 Starting Epoch 2
14   Epoch 2/20 | Batch 0 | Loss: 1.6511
15 ==> Epoch 2 Train Complete | Average BCE Loss: 2.0445
16  Validation Phase | Avg Val Loss: 1.3472
17  □ New best Tox21 model saved! (Lowest Val Loss: 1.3472)
18
19 Starting Epoch 3
20   Epoch 3/20 | Batch 0 | Loss: 0.9445
21 ==> Epoch 3 Train Complete | Average BCE Loss: 1.4343
22  Validation Phase | Avg Val Loss: 1.2867
23  □ New best Tox21 model saved! (Lowest Val Loss: 1.2867)
24
25 Starting Epoch 4
26   Epoch 4/20 | Batch 0 | Loss: 1.0894
27 ==> Epoch 4 Train Complete | Average BCE Loss: 1.2905
28  Validation Phase | Avg Val Loss: 1.2889
29  Model did not improve. Best Val loss remains: 1.2867
30
31 Starting Epoch 5
32   Epoch 5/20 | Batch 0 | Loss: 1.1734
33 ==> Epoch 5 Train Complete | Average BCE Loss: 1.2890
34  Validation Phase | Avg Val Loss: 1.2563
35  □ New best Tox21 model saved! (Lowest Val Loss: 1.2563)
36
37 Starting Epoch 6
38   Epoch 6/20 | Batch 0 | Loss: 1.6003
39 ==> Epoch 6 Train Complete | Average BCE Loss: 1.2709
40  Validation Phase | Avg Val Loss: 1.2582
41  Model did not improve. Best Val loss remains: 1.2563
42
```

43 Starting Epoch 7
44 Epoch 7/20 | Batch 0 | Loss: 1.4123
45 ==> Epoch 7 Train Complete | Average BCE Loss: 1.2652
46 Validation Phase | Avg Val Loss: 1.2594
47 Model did not improve. Best Val loss remains: 1.2563
48
49 Starting Epoch 8
50 Epoch 8/20 | Batch 0 | Loss: 1.1108
51 ==> Epoch 8 Train Complete | Average BCE Loss: 1.2588
52 Validation Phase | Avg Val Loss: 1.2531
53 ☐ New best Tox21 model saved! (Lowest Val Loss: 1.2531)
54
55 Starting Epoch 9
56 Epoch 9/20 | Batch 0 | Loss: 1.4384
57 ==> Epoch 9 Train Complete | Average BCE Loss: 1.2561
58 Validation Phase | Avg Val Loss: 1.2512
59 ☐ New best Tox21 model saved! (Lowest Val Loss: 1.2512)
60
61 Starting Epoch 10
62 Epoch 10/20 | Batch 0 | Loss: 1.1649
63 ==> Epoch 10 Train Complete | Average BCE Loss: 1.2494
64 Validation Phase | Avg Val Loss: 1.2324
65 ☐ New best Tox21 model saved! (Lowest Val Loss: 1.2324)
66
67 Starting Epoch 11
68 Epoch 11/20 | Batch 0 | Loss: 1.1028
69 ==> Epoch 11 Train Complete | Average BCE Loss: 1.2440
70 Validation Phase | Avg Val Loss: 1.2268
71 ☐ New best Tox21 model saved! (Lowest Val Loss: 1.2268)
72
73 Starting Epoch 12
74 Epoch 12/20 | Batch 0 | Loss: 0.9126
75 ==> Epoch 12 Train Complete | Average BCE Loss: 1.2368
76 Validation Phase | Avg Val Loss: 1.2253
77 ☐ New best Tox21 model saved! (Lowest Val Loss: 1.2253)
78
79 Starting Epoch 13
80 Epoch 13/20 | Batch 0 | Loss: 0.9541
81 ==> Epoch 13 Train Complete | Average BCE Loss: 1.2339
82 Validation Phase | Avg Val Loss: 1.2331
83 Model did not improve. Best Val loss remains: 1.2253
84
85 Starting Epoch 14
86 Epoch 14/20 | Batch 0 | Loss: 1.7996

```
87 ==> Epoch 14 Train Complete | Average BCE Loss: 1.2278
88   Validation Phase | Avg Val Loss: 1.2216
89   □ New best Tox21 model saved! (Lowest Val Loss: 1.2216)
90
91 Starting Epoch 15
92   Epoch 15/20 | Batch 0 | Loss: 1.4753
93 ==> Epoch 15 Train Complete | Average BCE Loss: 1.2246
94   Validation Phase | Avg Val Loss: 1.2113
95   □ New best Tox21 model saved! (Lowest Val Loss: 1.2113)
96
97 Starting Epoch 16
98   Epoch 16/20 | Batch 0 | Loss: 1.0769
99 ==> Epoch 16 Train Complete | Average BCE Loss: 1.2209
100  Validation Phase | Avg Val Loss: 1.2226
101 Model did not improve. Best Val loss remains: 1.2113
102
103 Starting Epoch 17
104   Epoch 17/20 | Batch 0 | Loss: 1.1519
105 ==> Epoch 17 Train Complete | Average BCE Loss: 1.2177
106   Validation Phase | Avg Val Loss: 1.2165
107 Model did not improve. Best Val loss remains: 1.2113
108
109 Starting Epoch 18
110   Epoch 18/20 | Batch 0 | Loss: 0.9409
111 ==> Epoch 18 Train Complete | Average BCE Loss: 1.2121
112   Validation Phase | Avg Val Loss: 1.2297
113 Model did not improve. Best Val loss remains: 1.2113
114
115 Starting Epoch 19
116   Epoch 19/20 | Batch 0 | Loss: 1.2312
117 ==> Epoch 19 Train Complete | Average BCE Loss: 1.2134
118   Validation Phase | Avg Val Loss: 1.2147
119 Model did not improve. Best Val loss remains: 1.2113
120
121 Starting Epoch 20
122   Epoch 20/20 | Batch 0 | Loss: 1.1180
123 ==> Epoch 20 Train Complete | Average BCE Loss: 1.2107
124   Validation Phase | Avg Val Loss: 1.2057
125   □ New best Tox21 model saved! (Lowest Val Loss: 1.2057)
126
127 Time: 0.24 minutes.
128
129 Process finished with exit code 0
130
```